interCloudNegotiation

Service Description

Service ID: *"interCloudNegotiation"*

Abstract
This document describes the `interCloudNegotiation` service, version 5.2.2, produced by the Gatekeeper support system of Arrowhead (documentation edition 5.3.0).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `interCloudNegotiation` service, which provides negotiation of the use of one provider in a neighbour cloud and opening of the GateTunnel session through which the consumer calls it. It is produced by the Gatekeeper system [1] in service version 5.2.2.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

2.1 Significant Prior Art

Successor of the Arrowhead 4.6 Gatekeeper inter-cloud negotiation (ICN, `/init_icn`).

2.2 How This Service Is Meant to Be Used

DynamicServiceOrchestration calls `negotiate` with the target cloud, provider, service and operations. The provider cloud's Gatekeeper checks authorization and opens a provider session at its GateTunnel; the local Gatekeeper opens the consumer session at the local GateTunnel and returns its endpoint as the provider interface.

2.3 Important Delimitations and Dependencies

Only the negotiated operations (method and path) can be used through the session. An AH-GT-1 (end-to-end) peer is refused rather than downgraded.

3 Service Interface

This section describes the interfaces to the `interCloudNegotiation` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.



Figure 1: SysML-style block of the `interCloudNegotiation` service and its operations.

The following interface operations are available.

3.1 operation negotiate (InterCloudNegotiationRequest) : InterCloudNegotiation Result

Negotiates one provider in a target cloud and returns the local session interface.

4 Information Model

Here, all data objects that can be part of the `interCloudNegotiation` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.3, which are used to represent things like hashes and identifiers.

4.1 struct InterCloudNegotiationRequest

Field	Type	Description
targetCloud	String	Provider cloud, `<Cloud>` or `<Cloud>|<Organization>` (alias `providerCloud`).
providerName	Name	Provider system.
serviceDefinition	Name	Service definition.
operations	List<String>	Operations to negotiate; default all offered.
consumerSystem	String	Consumer; default the requester.

4.2 struct InterCloudNegotiationResult

Field	Type	Description
negotiated	Boolean	Whether a session was opened.
reason	String	Refusal reason (when not negotiated).
providerCloud	String	Provider cloud, `<Cloud>|<Organization>`.
providerName	Name	Provider system.
serviceDefinition	Name	Service definition.
version	Version	Service version.
metadata	Metadata	Instance metadata.
sessionId	Identifier	GateTunnel session.
interface	Map<String, Object>	`generic_https` interface of the local GateTunnel session endpoint (`accessAddresses`, `accessPort`, `basePath`, `operations`).

4.3 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
Boolean	One out of `true` or `false`.
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Map	A set of key-value pairs.
Metadata	A set of key-value pairs with arbitrary, possibly nested values.
Name	A string identifier that is intended to be both human and machine-readable.
Object	Any structured value.
String	A chain of characters.
Version	Specifies a version, semantic versioning `major.minor.patch`.

5 References

[1] J. Delsing, “Gatekeeper – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-Gatekeeper-java-spring/documentation/v5.3.0/Gatekeeper-v5.3.0-SysD.pdf.

[2] ——, “interCloudNegotiation HTTP/TLS/JSON – Interface Design Description (Gatekeeper),” Eclipse Arrowhead, IDD 5.3.0, 2026, gatekeeper-v5.3.0-interCloudNegotiation-HTTP-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)
3	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	